

ARTICLE

High-throughput production of liver parenchymal microtissues and enrichment of organ-specific functions in gelatin methacrylamide microenvironment

Ramesh Pai Roopesh¹  | Senthilkumar Muthusamy¹  | Shiny Velayudhan²  |
Arumugham Sabareeswaran³  | Pallickaveedu RajanAsari Anil Kumar¹ 

¹Division of Tissue Culture, Department of Applied Biology, Biomedical Technology Wing, Sree Chitra Tirunal Institute for Medical Sciences and Technology, Thiruvananthapuram, Kerala, India

²Division of Dental Products, Department of Biomaterial Science and Technology, Biomedical Technology Wing, Sree Chitra Tirunal Institute for Medical Sciences and Technology, Thiruvananthapuram, Kerala, India

³Division of Experimental Pathology, Department of Applied Biology, Biomedical Technology Wing, Sree Chitra Tirunal Institute for Medical Sciences and Technology, Thiruvananthapuram, Kerala, India

Correspondence

Pallickaveedu RajanAsari Anil Kumar, Division of Tissue Culture, Department of Applied Biology, Biomedical Technology Wing, Sree Chitra Tirunal Institute for Medical Sciences and Technology, Poojapura, Thiruvananthapuram - 695 012, Kerala, India. Email: anilkumarpr@sctimst.ac.in

Funding information

Department of Biotechnology, Ministry of Science and Technology, Grant/Award Number: BT/RLF/Re-entry/48/2016; Department of Science and Technology, Ministry of Science and Technology, Grant/Award Number: TRC/8141/PSN

Abstract

Liver parenchymal microtissues (LPMTs) are three-dimensional (3D) aggregates of hepatocytes that recapitulate in vivo-like cellular assembly. They are considered as a valuable model to study drug metabolism, disease biology, and serve as ideal building blocks for liver tissue engineering. However, their integration into the mainstream drug screening process has been hindered due to the lack of simple, rapid techniques to produce a large number of uniform microtissues and preserve their structural-functional integrity over the long term. Here, we present a high-throughput methodology to produce LPMTs in a novel, economic, and reusable Hanging-drop Culture Chamber (HdCC). A drop-on-demand bioprinting approach was optimized to generate droplets of HepG2 cell suspension on a polyethylene terephthalate substrate. The substrates carrying droplets were placed inside a novel HdCC and incubated to obtain 1600 LPMTs having a size of 200–300 µm. Tissue size, cell viability, cellular arrangement and polarity, and insulin-mediated glucose uptake by LPMTs were analyzed. The microtissues were viable and exhibited an active response to insulin stimulation. Cells within the microtissue reorganized to form hepatic plate-like structures and expressed apical (Multidrug Resistance Protein 2 [MRP2]) and epithelial (Zonula Occludens 1 [ZO1]) markers. Further to maintain the structural integrity and enhance the functional capabilities, LPMTs were sandwiched within gelatin methacrylamide (GelMA) hydrogel and the liver-specific functions were monitored for 2 weeks. The results showed that the 3D structure of LPMTs in GelMA sandwich was maintained while the albumin secretion, urea synthesis, and cytochrome P450 activity were enhanced compared with LPMTs in suspension. In conclusion, this study presents a novel culture chamber for mass production of microtissues and a method for enhancing organ-specific functions of LPMTs in vitro.

KEYWORDS

bioprinting, gelatin methacrylamide, hanging-drop culture, microtissues, sandwich culture

1 | INTRODUCTION

The liver is the largest gland in the body, performing a plethora of endocrine and exocrine functions. As a major organ involved in xenobiotic metabolism, it is highly susceptible to drug-induced liver injury (DILI). Therefore, it is critical to identify the toxicity of new chemical entities during drug development. However, despite having strict evaluation strategies in the preclinical phase, DILI remains the leading cause of drug failures in the clinical trials and postmarketing withdrawals (Watkins, 2011). The limitations of conventional hepatocyte monolayer culture to express mature liver phenotype and the interspecies difference in enzyme-mediated drug metabolism related to animal models are the primary cause of unprecedented complications due to DILI (Martignoni et al., 2006). Meanwhile, the global burden of liver diseases accounts for approximately two million deaths per year and liver transplantation is the only treatment strategy for end-stage liver diseases (Asrani et al., 2019). Hence, alternative physiologically relevant human liver models are needed for toxicity testing and engineering liver tissue grafts. Of late, three-dimensional (3D) liver models such as hepatocyte spheroids or microtissues, liver organoids, and 3D bioprinted liver constructs have exhibited enhanced efficiency in predicting human DILI and restoring functions in liver failure models (Jin et al., 2018; Nguyen et al., 2016; Ramaiahgari et al., 2014; Yang et al., 2021). Among these, hepatocyte microtissues are easy to prepare and amenable for high-throughput analysis (Hiemstra et al., 2019). Hepatocyte microtissues are self-organized aggregates of hepatocytes that exhibit complex cell-to-cell interaction and hepatic polarity, and they develop physiological gradients within the tissue (Glicklis et al., 2004).

Methods to generate microtissues can be broadly classified into two: scaffold-based and scaffold-free methods. Scaffold-based methods include culturing cells on scaffolds synthesized from natural polymers (agarose, collagen, etc.) or synthetic polymers (PEG, NIPAM, PLA, etc.; Lee, Kim, et al., 2015; Liao et al., 2019; Polonio-Alcalá et al., 2019; Wang et al., 2011). Scaffold-free methods include hanging-drop culture, culture in nonadherent wells, spinner flask, rotary flask, magnetic levitation, and microfluidics (Haisler et al., 2013; Manley & Lelkes, 2006; Okuyama et al., 2010; Shri et al., 2017). Among these, hanging-drop culture is the most common and straightforward method for generating microtissues. In this method, a cell suspension is transferred to a substrate as droplets, and the substrate is inverted to hang the suspension under gravity. The gravitational force induces cell aggregation and microtissue formation. Although microtissues of defined size, cell number, and composition can be generated using this method, manual dispensing of cells is extremely labor intensive. Recently use of automated liquid handling systems along with multiwell hanging-drop microplates has addressed the limitations to some extent (Drewitz et al., 2011). But at the same time, the microplates are expensive and the restriction in the number of microtissues produced per plate limits their use at the industrial scale. Apart from these limitations, preserving the structural integrity of microtissues during *in vitro* culture is important to maintain its differentiated functions (Chang & Hughes-Fulford, 2009).

Earlier studies have shown that the sandwich culture of hepatocyte monolayer preserved hepatic polarity and improved differentiated functions (Berthiaume et al., 1996; Du et al., 2008). However, such a 3D matrix configuration has not been yet evaluated for the culture of hepatocyte microtissues.

To address these shortcomings, we developed a cost-effective and high-throughput method to produce hepatocyte microtissues, hereafter referred to as liver parenchymal microtissues (LPMTs). In this methodology, we used microvalve-based bioprinting technique to generate a high density of low-volume cell-laden droplets on polyethylene terephthalate (PET) substrate. Multiple substrates were then placed inside a novel, economic, and reusable Hanging-drop Culture Chamber (HdCC) to form microtissues. High efficiency of LPMT formation and harvesting was achieved using our culture platform. An in-depth characterization of LPMTs revealed that the microtissues were structurally and functionally mature. Further, we explored the potential of the gelatin methacrylamide (GelMA) hydrogel sandwich system to enhance the differentiated functions of LPMTs during *in vitro* culture.

2 | MATERIALS AND METHODS

2.1 | Culture of hepatocytes

Human hepatocellular carcinoma cell line, HepG2 (National Center for Cell Science), was cultured in Minimum Essential Medium (MEM, Merck-Sigma, Cat.# M0769) with 10% Fetal Bovine Serum (FBS, Gibco, Cat.#10270106) and 0.1% Penicillin-Streptomycin antibiotic (Pen-Strep, Gibco, Cat.#15070-063). Cells were incubated in CO₂ incubator (New Brunswick, Galaxy 170S) set at 37°C, 5% CO₂ and >90% relative humidity with medium change on every alternate day.

2.2 | Design and fabrication of an HdCC

Design of the HdCC was prepared using an open-source general-purpose parametric 3D computer-aided design (CAD) modeler (FreeCAD Ver 0.18). Clear transparent acrylic sheets of 6 mm thickness were cut using a laser cutter (Han's Yueming; 100 V) to pre-defined dimensions. The key internal component of HdCC is a base plate on which three vertically placed pillars fixed in position with a top plate. The vertical pillar has grooves of 2 × 4 mm size on the inner side that is accurately positioned to adjacent pillars. The vertical pillars can accommodate eight substrates supporting hanging drops.

2.3 | Optimization of microvalve-based printing

Microvalve-based printing allows a user to generate arrays of droplets in a high-throughput manner. We used an electromagnetic dispensing valve (regenHu, Cat.#MVC-D0.3S0.1) having an inner diameter of 300 and 100 µm stroke to dispense the droplets.

The microvalve was connected to a cartridge containing water and loaded into a cell-friendly printhead (regenHU, Cat.#CF-300N) of the bioprinter 3D Discovery Evolution (regenHU). The template to print a droplet array (15 × 14 droplets) was created in BioCAD 1.1 version software (regenHU). The G-Code was generated from the CAD design and imported into the machine control software (HMI regenHU). A PET sheet (Model#CT100, Desmat) of 70 × 65 mm size and a thickness of 100 µm were used as the substrate for hanging drop. The optimum diameter of the sitting droplet was calculated to be 2 mm, and a pitch of 1.4 mm between the droplets was required to accommodate a 15 × 14 droplet array within the print area. However, the diameter of the droplet depends upon the volume of the dispensed droplet. The volume dispensed by the microvalve, in turn, depends upon the printhead parameters, such as valve opening time (VOT), valve closing time (VCT), and the cartridge pressure (CPR). To determine the optimum printhead parameters multiple arrays of water droplets were printed at a travel speed of 12 mm/s with variable VOT (10, 15, 20, 25, and 30 s) and CPR (2, 4, 6, 8, and 10 kPa) while keeping the VCT constant at 10 milliseconds. The diameter and the weight of randomly selected droplet were measured to estimate the volume (1 mg = 1 µl).

The volume of the printed drops was analyzed indirectly from the diameter of the printed drops measured using ImageJ software. The droplet array was divided into top, middle, and bottom parts each section having 14 × 5 droplets. Random images from each section were captured using a 3D stereomicroscope (Leica S8 Apo) and the feret diameter of the drops were measured. Further, the approximate volume of printed drop was estimated by comparing with liquid drops of different volumes (0.5, 1.5, 2.5, 3.5, and 4.5 µl) manually dispensed using a micropipette. The diameter of the printed drops was and compared with manually dispensed drops.

2.4 | Microvalve-based drop-on-demand bioprinting of hepatocytes

HepG2 cells from a confluent culture were harvested using 0.25% trypsin-ethylenediaminetetraacetic acid (EDTA) solution (Gibco, Cat.#25200072) and different homogeneous cell suspensions containing cell concentration (2×10^5 , 4×10^5 , and 10×10^5 cells/ml) were prepared in MEM. The cell suspension was loaded into a 3 ml cartridge assembled on the bioprinter head. A motorized cell agitator was inserted into the cell suspension in the cartridge, and the cell suspension was agitated at the lowest speed for 5 s. A droplet array containing 15 × 14 droplets was bioprinted on the PET culture substrate at 8 kPa CPR, 20 ms VOT, and 10 ms VCT. The substrate was inverted carefully without disturbing the droplets and placed inside the HdCC. Bioprinted substrates were placed inside multiple HdCC with each unit having eight substrates each. The HdCC was closed and incubated inside CO₂ incubator for 72 h. The accuracy and the high-throughput nature of microvalve-based bioprinting were compared with the manually dispensing method and their culture in HdCC. In the manual method, 10 µl of cell suspension containing 500

cells was dispensed ($n = 200$) using a micropipette on PET sheets, was inverted and incubated inside the HdCC at 37°C for 72 h.

The efficiency of microtissue formation in both methods was determined by examining entire droplets from three different sheets under a phase-contrast microscope (Nikon TS100). The efficiency of microtissue generation was calculated using the equation given below and the results were presented as mean ± standard deviation.

$$\text{Efficiency of microtissue generation (\%)} = \frac{\text{Total no. of microtissues formed}}{\text{Total no. of drops dispensed}} \times 100.$$

2.4.1 | LPMT harvesting, culture, and size analysis

The microtissues from each PET substrate (drop number = 210) were gently washed out using culture medium (1 ml) and allowed to flow into a centrifuge tube. The microtissue suspension was centrifuged at 11g for 30 s. The supernatant was aspirated and 1 ml fresh culture medium was added to the pellet. The suspension containing microtissues were transferred to a 12-well, nontreated plate (Eppendorf, Cat.#0030721012) and maintained inside a CO₂ incubator. Phase-contrast images of the LPMT ($n = 40$) were captured using a digital camera (VUE, Model# US300) connected to the light microscope (Nikon Eclipse TS2-FL). The feret diameter of the microtissues was measured using ImageJ analysis software.

2.4.2 | Assessment of cell viability

Viability of LPMT was assessed on the 7th day by live–dead staining. Microtissues were pelleted by centrifuging and dispersed in serum-free medium containing 10 µg/ml fluorescein diacetate (FDA, Sigma Cat# F7378) and 0.05 µg/ml propidium iodide (PI; HiMedia, Cat#TC252). After incubation for 2 min in the dark, the stain was removed, and LPMT was rinsed using fresh serum-free MEM. The viability was monitored and imaged using a confocal laser scanning microscope (Olympus, Model# FV3000).

Adenosine TriPhosphate (ATP) content is a direct indicator of metabolic activity in viable cells. The ATP content of LPMTs was quantified using CellTiter-Glo 3D cell viability assay kit (Promega, Cat.#G9681) as per the product instructions. Around 200 LPMTs ($n = 3$) cultured for days 1, 7, and 15 were analyzed for measuring ATP. Briefly, LPMTs were collected, pelleted, and resuspended in 100 µl of culture medium. The suspension was transferred to a white opaque 96-well microplate (Thermo Scientific, Cat.#9502887) in triplicates. The plate was equilibrated to room temperature (25°C) for 25 min, and ATP assay reagent (100 µl) was added to each well. The plate was shaken vigorously on a microplate orbital shaker at 1000 rpm for 10 min and was then incubated for 20 min at room temperature. The luminescence intensity was measured using a Multimode plate reader (Plate Chameleon V 425-106, Hidex). Percentage of viable cells was calculated by normalizing the luminescence values at different time points to the value at the beginning of the culture.

2.4.3 | Protein estimation

The total protein content of LPMTs was quantified using the BCA protein assay kit (TaKaRa Cat.#T9300A) as per manufacturer instructions. Briefly, about 200 microtissues were lysed in a RIPA lysis buffer (500 µl; Origin, Cat.#PB103-01). Before lysis, the LPMTs from sandwich-cultured microtissues were isolated by collagenase A treatment. The standard solution was prepared by linear dilutions of Bovine Serum Albumin (BSA) dissolved in the lysis buffer. The tissue lysate and the standard dilutions were transferred to a 96-well plate, and an equal volume of BCA working solution was added to each well. The content was mixed thoroughly by shaking using a microplate orbital shaker (MixMate 5353, Eppendorf) at 1000 rpm for 5 min, followed by incubation at 37°C for 1 h. The optical density was quantified by measuring absorbance at 562 nm using a microplate spectrophotometer (Power Wave HT, BioTek). The concentration of protein was calculated from a standard curve.

2.4.4 | Glucose uptake assay

Insulin-stimulated glucose uptake by LPMTs was estimated using fluorescent-tagged 2-deoxy-glucose analog (2-deoxy-2-[(7-nitro-2,1,3-benzoxadiazol-4-yl)amino]-D-glucose, 2NDBG; Abcam Cat. #146200). Briefly, 210 LPMTs ($n = 3$) were collected and pelleted by centrifugation at 11g for 30 s. The pellet was rinsed with phosphate-buffered saline (PBS) and again pelleted before resuspending in glucose-free DMEM (100 µl; Himedia, Cat.#AT186). The LPMT suspension was transferred to a clear-bottom black 96-well microplate (Thermo Scientific Cat.#10281092) and incubated for 2 h at 37°C for starving. The LPMTs were then treated with 0 or 32 IU/ml insulin (Cat.#HI0240, Huminsulin R, Eli Lilly Company) in DMEM containing 2-[N-(7-nitrobenz-2-oxa-1,3-diazol-4-yl) amino]-2-deoxy-D-Glucose (200 µg/ml) for 2 h at 37°C. The LPMTs were washed twice with PBS with intermediate centrifugation using a microplate centrifuge (Universal 32R, Hettich). Microtissues were there suspended in PBS (100 µl), and fluorescence intensity was measured using a multimode microplate reader (BioTek Synergy H1) at an Ex/Em wavelength of 485/535 nm, respectively. The total protein of LPMTs was estimated using the BCA Protein Assay Kit (Takara, Cat.#T9300A). The glucose uptake by LPMTs was determined from a standard curve for 2-NBD-Glucose and was normalized to the total cellular protein content.

2.4.5 | Histomorphology analysis and immunostaining

LPMTs after 7 days of culture were embedded in 2% agarose solution prewarmed to 37°C. The suspension was transferred to a 96-well plate and allowed to cool to room temperature for gelation. The agarose embedded LPMTs were then fixed in 10% neutral buffered formalin for 24 h. The samples were dehydrated and processed for

paraffin embedding. Embedded samples were cut into 5 µm thick sections using a microtome (Leica RM2245) and mounted on charged glass slides. The sections were stained with haematoxylin and eosin stain. The stained sections were observed under the bright-field mode of a light microscope (Eclipse TS100, Nikon), and digital images were captured (Digital Sight camera # DS-Fi2, Nikon).

Paraffin sections of LPMTs were used for immunofluorescence staining. The sections were dewaxed at 60°C, cleared by two xylene washes and then rehydrated using decreasing concentrations of isopropanol (90% and 75%) and tap water. Antigen retrieval was performed at 60°C in sodium citrate buffer (pH = 6). The sections were permeabilized in 0.025% Triton X-100, nonspecific binding sites were blocked using 1% BSA. The sections were incubated with primary antibodies against MRP2 (Abcam, Cat.#ab15603, 1:20 dilution) and ZO1 (Abcam, Cat.#ab214228, 1:250 dilution). They were detected using a secondary antibody tagged with the fluorescent probe Alexa 488. The cell nucleus was counterstained with PI, and sections were mounted using PBS:glycerol (1:1) solution. The samples were then observed under differential interference contrast mode, and fluorescence images were captured using a confocal laser scanning microscope (Nikon).

2.5 | LPMT sandwich culture

2.5.1 | GelMA synthesis

GelMA was synthesized, as previously reported (Lee, Shirahama, et al., 2015). Briefly, 10 g of Type A porcine skin gelatin (Merck-Sigma, Cat.#G-2625) dissolved in 100 ml of 0.1 M carbonate buffer (pH-10.5) and 1 ml methacrylic anhydride (MAA; Merck-Sigma, Cat. #276685) was added stepwise. The solution was stirred at 50°C for 3 h while keeping the pH stable at 10.5. The GelMA thus formed was dialyzed against water using 12–14 kDa cutoff dialysis tubing (Merck-Sigma, Cat. # D9652) for 72 h at 40°C. The dialyzed GelMA was lyophilized, sterilized by ethylene oxide (ETO) gas and stored at –20°C until use.

2.5.2 | Characterization of GelMA

Nuclear magnetic resonance (NMR) spectroscopy

The functionalization of gelatin by the methacrylamide group was verified by ¹H NMR spectroscopy. Five milligrams of gelatin and GelMA polymer was dissolved separately in deuterium oxide (D₂O) at 40°C, and their spectrum was recorded in a Fourier Transform-NMR spectrometer (Bruker, Model# Avance 400) at 400 MHz using tetramethylsilane as the internal standard.

Scanning electron microscopy

The surface morphology of lyophilized GelMA was visualized using a scanning electron microscope (SEM). Cylindrical hydrogel samples having 8 mm diameter and 2 mm height were lyophilized for 48 h and

sputter-coated with gold (HITACHI, E1010). The images were then captured using an SEM (HITACHI, S-2400) set at 15 kV.

Microcomputed tomography (micro-CT)

The porosity of lyophilized GelMA was analyzed using a micro-CT scanner (Scano Medical, Model# μ CT 40). Cylindrical hydrogel samples having 12 mm diameter and 6 mm height were cast and lyophilized for 48 h. Lyophilized samples were scanned using a cone-beam X-ray source set at 70 kV and 160 μ A having a 5 μ m spot size. Three-dimensional reconstruction of the 2D images was done using the software μ CT V6.1. Morphometric analysis was carried out to determine polymer volume/total volume ratio (PV/TV) and pore size distribution. Percentage porosity of the hydrogel was calculated from Equation (1).

$$\text{Porosity (\%)} = 1 - \left(\frac{\text{PV}}{\text{TV}} \right) \times 100. \quad (1)$$

2.5.3 | Preparation of LPMT sandwich culture

The LPMTs were sandwiched between two layers of GelMA hydrogel (Figure 8a). The prepolymer solution of GelMA was prepared in serum-free MEM containing 10% (w/v) GelMA and 1% (w/v) Irgacure-2959 (2-hydroxy-4'-(2-hydroxyethoxy)-2-methylpropiophenone; Merck-Sigma, Cat.#410896) as photoinitiator. The first layer of hydrogel was prepared by pouring GelMA solution (100 μ l) into a cylindrical mold (5 mm diameter) and was crosslinked with ultraviolet (UV) light of 365 nm for 5–7 min. The hydrogel was placed inside a 12-well plate, and approximately 200 LPMTs ($n = 3$) were carefully transferred over the hydrogel. The plate was incubated inside a CO₂ incubator for 30 min for the LPMTs to settle on the hydrogel. The second layer of GelMA solution (25 μ l) was poured on the LPMTs and crosslinked as mentioned above. The enhancement of tissue-specific functions of LPMTs by GelMA was studied by comparing GelMA sandwich-cultured LPMTs (GelMA sw-LPMTs) with Alginate sandwich-cultured LPMTs (Alginate sw-LPMTs). A 5% (w/v) GelMA hydrogel sandwich was made for comparison using the same procedure as described above. The sodium alginate powder (Merck-Sigma, Cat.#A2033) was first sterilized by exposing it to UV radiation for 30 min within a biosafety cabinet, followed by pasteurization at 60°C for 30 min. Alginate was dissolved in serum-free MEM to prepare a 5% (w/v) solution. The first layer of alginate hydrogel was prepared in a cylindrical mold by crosslinking using 20 mM calcium chloride solution for 10 min. About 200 LPMTs ($n = 3$) were carefully transferred over the hydrogel layer and the second layer of alginate solution was added on settled LPMTs. The second layer of alginate was crosslinked to form the hydrogel sandwich, and 1 ml culture medium was added to each well. The sandwich culture was incubated at 37°C inside the humidified incubator with 5% CO₂ for 15 days.

2.5.4 | Functional analysis of sandwich-LPMTs

Spent media from suspension and sandwich-LPMT cultures were collected from triplicate samples at 48 h intervals for 15 days and stored at –80°C. The albumin and the urea secreted were estimated using Human Albumin ELISA kit (Bethyl Laboratories Inc., Cat.#E88-129) and Urea assay kit (Biochain, Cat.#Z5030016), respectively, as per the manufacturer's protocol. The absorbance values were measured using a microplate spectrophotometer (BioTek Model#). The albumin and urea values estimated from the assay were normalized to the total protein content of LPMTs.

CYP3A4 enzyme activity in LPMTs was determined using P450-Glo CYP3A4 assay kit (Promega, Cat.#V9001) as per manufacturers protocol. Briefly, LPMTs from 210 droplets ($n = 3$) were seeded into each well of a white opaque 96-well microplate. The microtissues were then incubated in 150 μ l of luciferin-Isopropyl Alcohol (IPA) substrate solution and incubated for 1 h at 37°C. The substrate solution was then replaced with 50 μ l of luciferin detection reagent, and the plate was incubated at room temperature for 20 min. The luminescence intensity was measured using a multimode plate reader (Plate Chameleon V, Hidex Model#425-106). For sandwich-cultured LPMTs, to retrieve the microtissues, GelMA sw-LPMTs were treated with collagenase A enzyme (Cat.#10103586001, Merck-Sigma,) at a final concentration of 1 mg/ml for 20 min and Alginate sw-LPMTs were treated with alginate dissolution solution (0.03 M EDTA, 0.055 M sodium citrate and 0.15 M sodium chloride in water, pH 7) for 1 min. The microtissues were then separated by centrifugation at 11g for 30 s and were seeded into the microplate for assay.

2.5.5 | Statistical analysis

Statistical analysis was performed with one-way ANOVA using the GraphPad Prism (Version 8), and all data were expressed as mean $\pm$ standard deviation. Student's *t* tests analyzed differences between groups, and *p* value of <0.05 was considered as significant.

3 | RESULTS

3.1 | Design and fabrication of an HdCC

The chamber was compact in size, transparent, easy to assemble, and portable (Figure 1). A single chamber can accommodate eight substrates having the dimensions 70 mm $\times$ 65 mm $\times$ 0.1 mm (length $\times$ breadth $\times$ height). The design ensured a closed environment and maintained an aseptic condition within the chamber. The chamber was reused after proper washing and ETO sterilization.

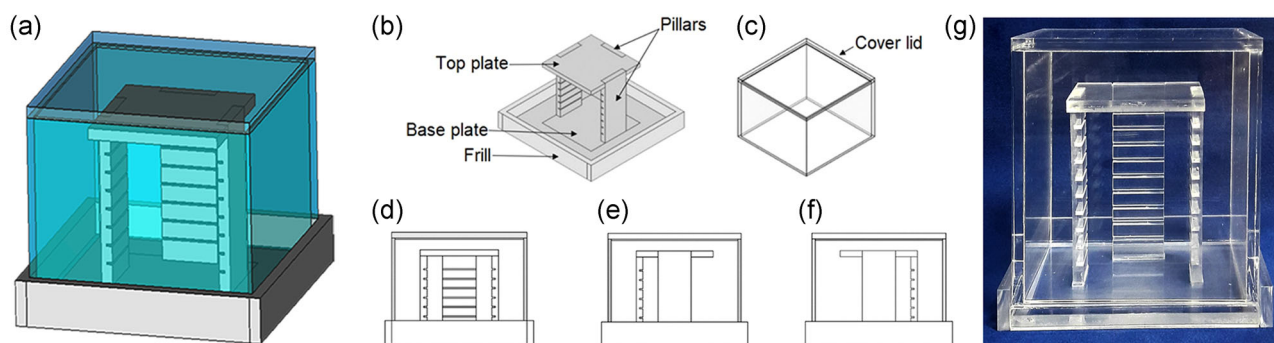


FIGURE 1 Design of the Hanging-drop Culture Chamber. (a) The computer-aided design (CAD) model of the HdCC. (b) Inner components, such as base plate, vertical pillars, and the top plate. (c) The lid covers the unit entirely for maintaining the aseptic environment around the inner components. Different views of the chamber: (d) front view, (e) left side view, and (f) right side view. (g) The image of HdCC used for the experiments. HdCC, Hanging-drop Culture Chamber

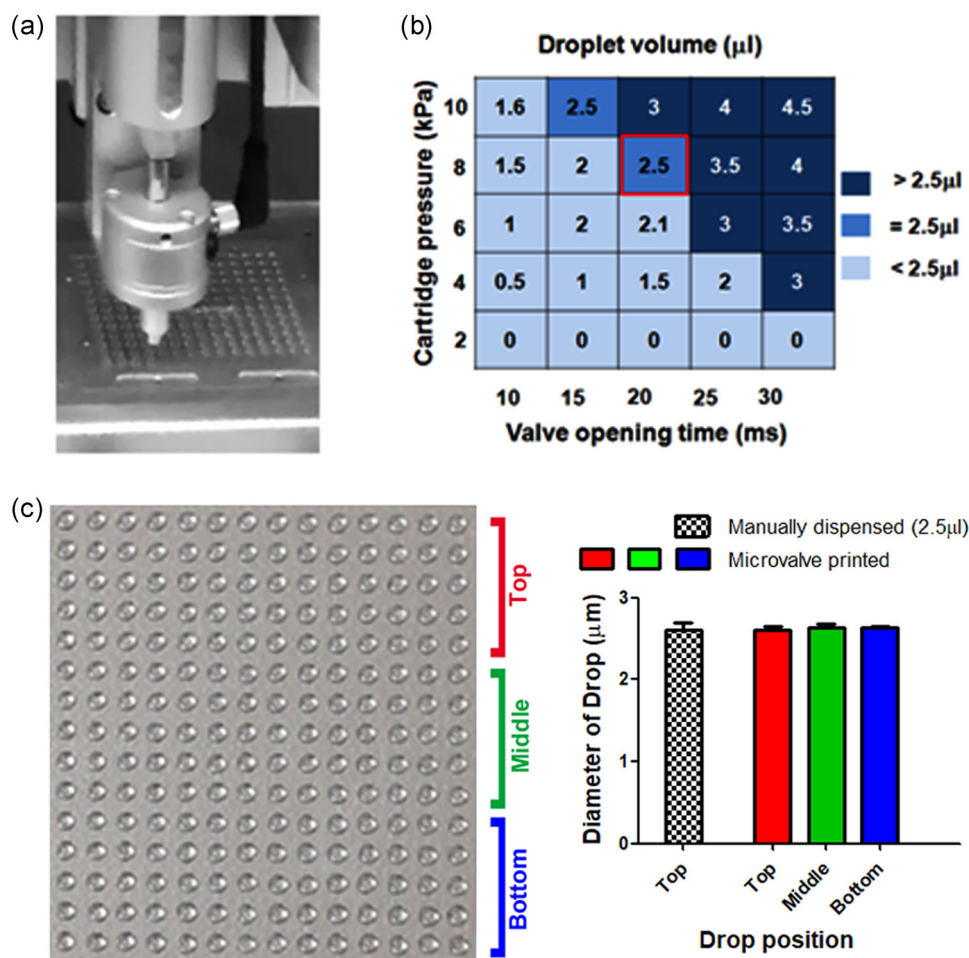


FIGURE 2 Determining the optimum printhead parameters for microvalve-based printing: (a) A digital image of the cell-friendly printhead with the cartridge connected to the microvalve. (b) Phase diagram depicting the variation in droplet volume with varying cartridge pressure and valve opening time. Values of the parameters corresponding to the highlighted cell (red) were considered optimum for microvalve-based printing. (c) A digital image showing a uniform droplet array of water printed using the optimum value and the uniform droplet size across different locations of the array (top, middle, and bottom) compared with manual dispensing. Scale bar = 4 mm

3.2 | Optimization of microvalve-based printing

Liquid droplets having a volume ranging from nanoliters to microliters were printed at high precision and optimum speed using a microvalve nozzle (Figure 2a). A droplet volume of 2.5 μl yielded the desired diameter close to 2 mm. The results from 25 different combinations of printhead parameters with VOT and CPR as variables were plotted into a matrix format as shown in Figure 2b. It was observed that the required volume of 2.5 μl was obtained at two conditions. The settings with a lower CPR of 8 kPa and 20 ms of VOT were selected for cell printing. Figure 2c shows the array of 15 $\times$ 14 droplets printed on a PET substrate. The time taken to complete a single array was 45 s. The droplets were uniform in size and shape, with adequate space between one another. The variation in size of printed droplets was analyzed using ImageJ software. The diameter of the printed droplet analyzed randomly from 5 $\times$ 14 droplets from top, middle, and bottom regions of array, matched with the manually dispensed drops having a volume of 2.5 μl . There was no difference in droplet size across different locations (top, middle, and bottom) of the array (Figure 1S).

3.3 | Microvalve-based drop-on-demand bioprinting of hepatocytes

A uniform 15 $\times$ 14 droplet array containing $\sim$ 500 cells/droplet was bioprinted on the PET substrate and cultured as a hanging drop inside the HdCC device (Figure 3a,b). Cells within the droplet formed an aggregate after 24 h of incubation at 37°C. The cell aggregate became more compact after 48 h and formed a spherical microtissue at the end of 72 h postprinting (Figure 3c). The efficiency of microtissue formation by microvalve-based bioprinting ($97.4\% \pm 0.7\%$) was significantly higher compared with the manual dispensing method ($62.8\% \pm 2\%$; Figure 3d). We observed that the size (diameter) of the microtissues generated on different PET substrates varied significantly during the initial stages of bioprinting. The nonuniformity among the microtissues was due to the difference in cell number, which occurred due to cell sedimentation during bioprinting. To overcome this limitation, we introduced a short agitation step before initiating each print command. Cell bioprinting following agitation yielded uniform microtissues. Figure 3e shows the variation in size of

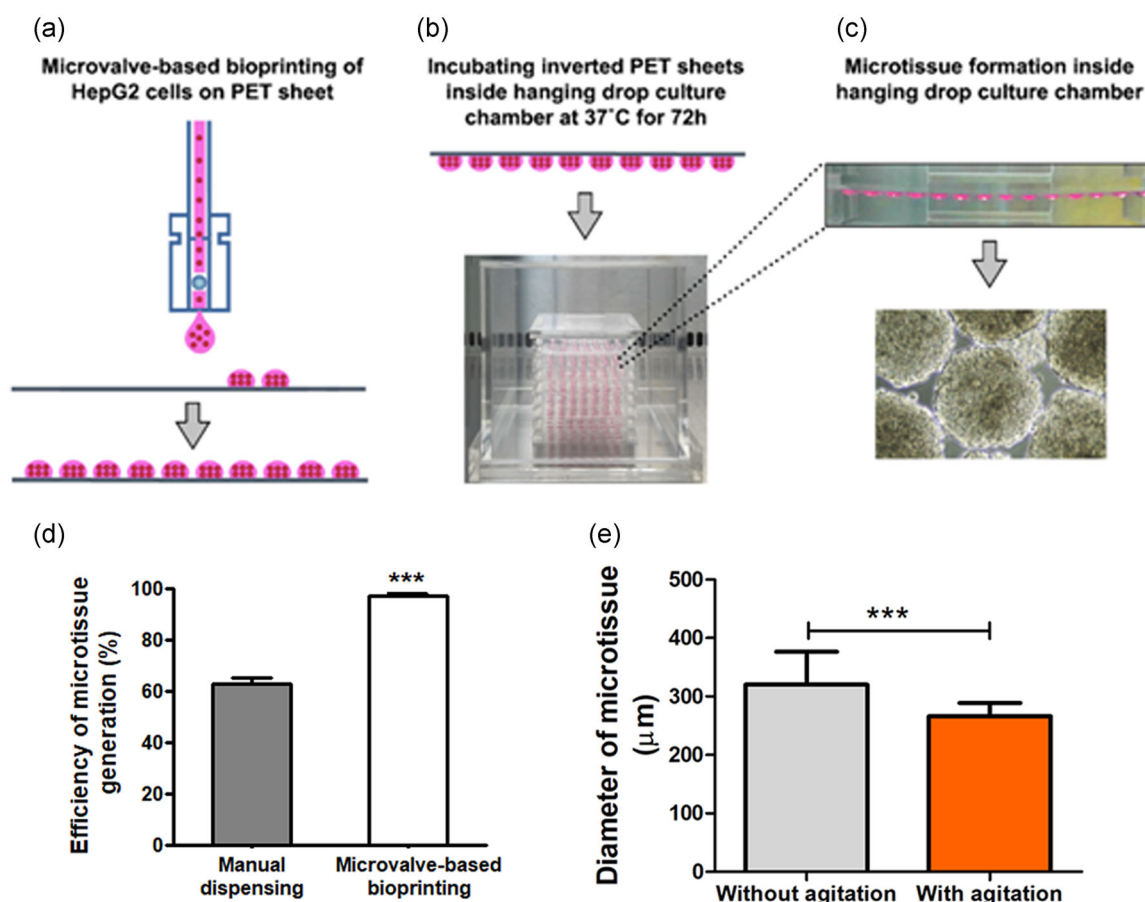


FIGURE 3 Steps in the generation of LPMTs in HdCC and efficacy of microvalve-based bioprinting. (a) Illustration showing microvalve-based bioprinting. (b) Culture of HepG2 cells in hanging drops inside HdCC. (c) Formation of microtissue inside HdCC. (d) Data showing the comparison between manual dispensing and microvalve-based bioprinting method for their efficiency in generating microtissues ($n = 630$, *** $p < 0.001$). (e) Data showing variation in size of the microtissues formed post-bioprinting with and without agitation. HdCC, Hanging-drop Culture Chamber; LPMT, liver parenchymal microtissue; PET, polyethylene terephthalate

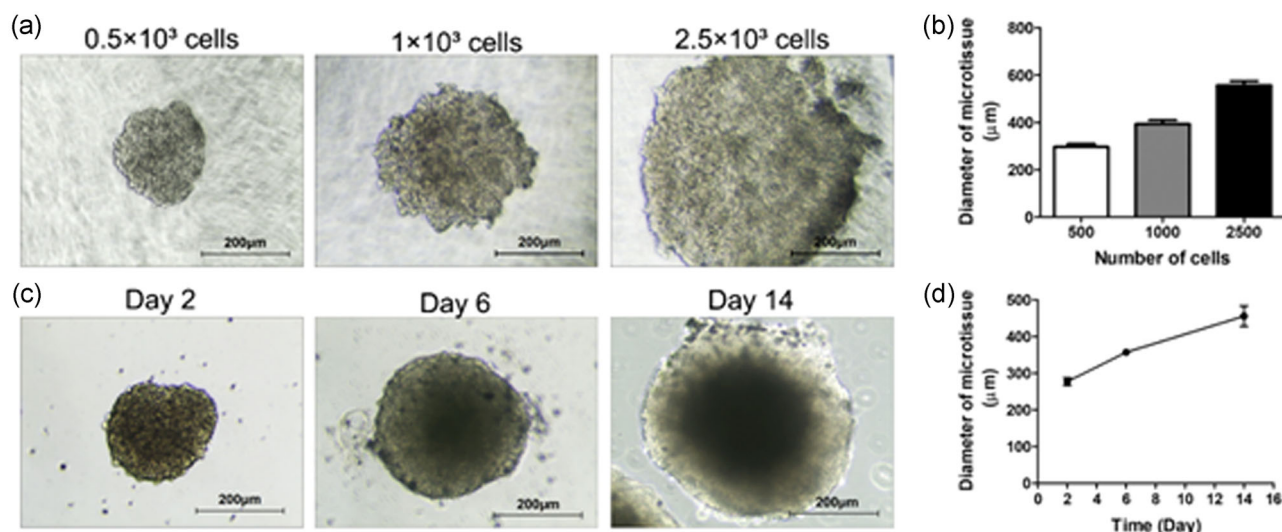


FIGURE 4 Assessing the size and stability of the LPMTs. (a) Phase-contrast images of representative microtissues generated from three different cell concentrations after 72 h. (b) Data representing the average diameter of the microtissues ($n = 4$) with different cell numbers after 72 h. (c) Phase-contrast images of the microtissue (500 cells) depicting the growth of the microtissue in culture. (d) The changes in the size of the microtissue (500 cells) over time in culture ($n = 9$). Scale bar = 200 μm . LPMT, liver parenchymal microtissue

the microtissues generated in the bioprinted droplets with and without agitation.

3.3.1 | Microtissue harvesting, culture, and size analysis

A single sheet of PET substrate generated around 200 microtissues, and about 1600 microtissues were obtained from an HdCC chamber. The efficiency in harvesting microtissue from PET sheet was $92.2\% \pm 3\%$. A positive correlation was observed between the number of cells dispensed per droplet and the size of the microtissue (Figure 4a). The average diameter versus cell number in the microtissue is shown in Figure 4b. The diameter of LPMT and the cell number per microtissue was 297.15 ± 11 , 393.69 ± 16 , and $557.71 \pm 17 \mu\text{m}$ for 0.5×10^3 , 1×10^3 , and 2.5×10^3 cells, respectively. Microtissues with the smallest diameter were chosen for the later experiments, considering the efficient nutrient transfer possibilities. The microtissues were stable without any sign of disintegration up to 2 weeks in suspension culture (Figure 4c). The microtissue diameter increased over time with an increment of 28.8% and 35.6% on days 6 and 14 of culture to the second day (Figure 4d). The increase in diameter of the microtissue could be attributed to the cell proliferation at the peripheral zone.

3.3.2 | Assessment of cell viability

Microvalve-based bioprinting of HepG2 cells yielded viable LPMTs. The total ATP content in microtissues was quantified to determine

the number of metabolically active cells. The microtissues exhibited a stable production of ATP up to 2 weeks in culture (Figure 5a). Percentage of cell metabolic activity at days 1, 7, and 15 was $100\% \pm 11\%$, $109\% \pm 20\%$, and $99.4\% \pm 2\%$, respectively. Further, we confirmed the cell viability of 7-day old LPMTs using live/dead staining and confocal microscopy imaging. The Z-stack projections (Figure 5b) of the stained LPMTs revealed that the cells were viable in the microtissue.

3.3.3 | Glucose uptake assay

Hepatic glucose uptake and release are dependent on the ability of microtissues to respond to insulin. The insulin sensitivity of LPMTs was assessed by quantifying the uptake of fluorescently labeled glucose by microtissues in the presence and absence of insulin (Figure 5c). The quantity of glucose uptake by microtissues in the presence of insulin was significantly higher ($646.6 \pm 22 \text{ ng}/\mu\text{g}$ total protein) compared with the untreated group ($505.8 \pm 5 \text{ ng}/\mu\text{g}$ total protein). The microtissues were intact, and labeled glucose uptake was visibly different between the two groups (Figure 5d).

3.3.4 | Histomorphology analysis and immunostaining

The structural characteristics of LPMTs were analyzed by staining paraffin-embedded sections. Hematoxylin and eosin (H&E) staining of cultured microtissue after 7 days (Figure 6a) showed dense tissue-like cellular distribution mimicking the hepatic plates

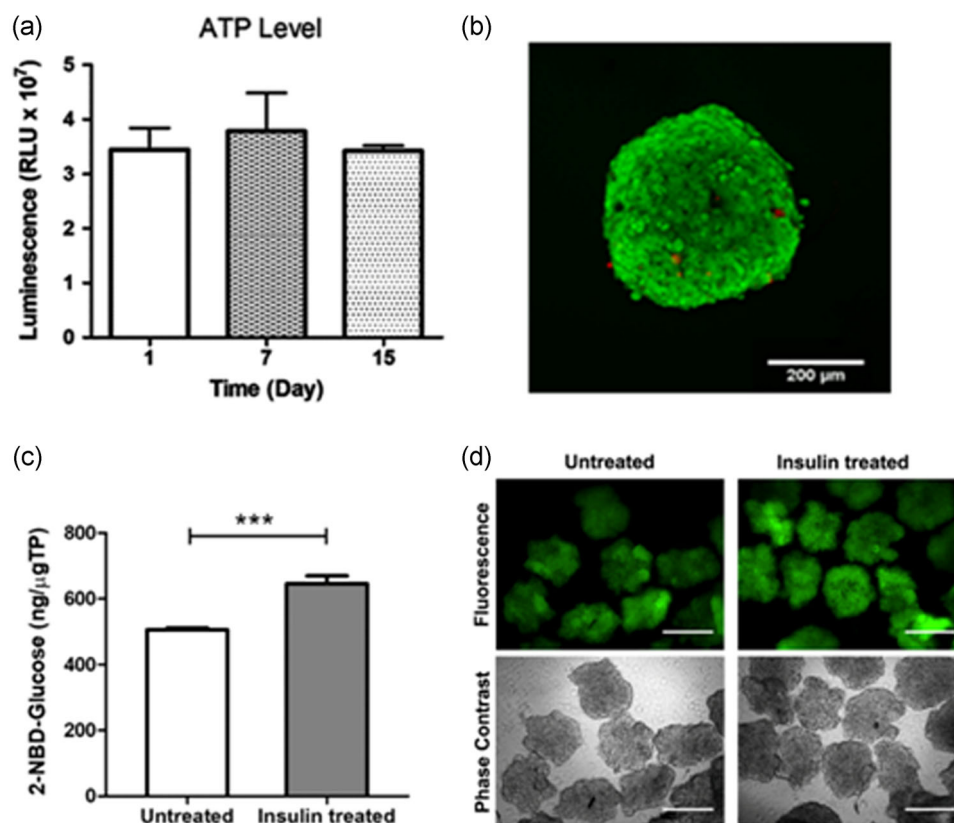


FIGURE 5 Cell viability and insulin-stimulated glucose uptake in LPMTs. (a) Results of cell viability assay calculated based on quantification of ATP in the tissue lysate. Relative Luminescence Unit (RLU) correlates with the ATP level and thus with the quantity of metabolically active (viable) cells (a pooled sample of 210 microtissues was analyzed in triplicates). (b) Confocal image showing live/dead staining of 7-day old LPMT with FDA (green, live cells) and PI (red, dead cells). (c) Data representing the quantity of 2-NBD glucose absorbed by microtissues in the absence and presence of insulin. (d) Fluorescent image showing uptake of 2-NBD glucose by microtissues and phase-contrast images showing the integrity of the microtissues. Scale bar = 200 μm. *** $p < 0.01$. ATP, Adenosine TriPhosphate; FDA, fluorescein diacetate; LPMT, liver parenchymal microtissue; PI, propidium iodide

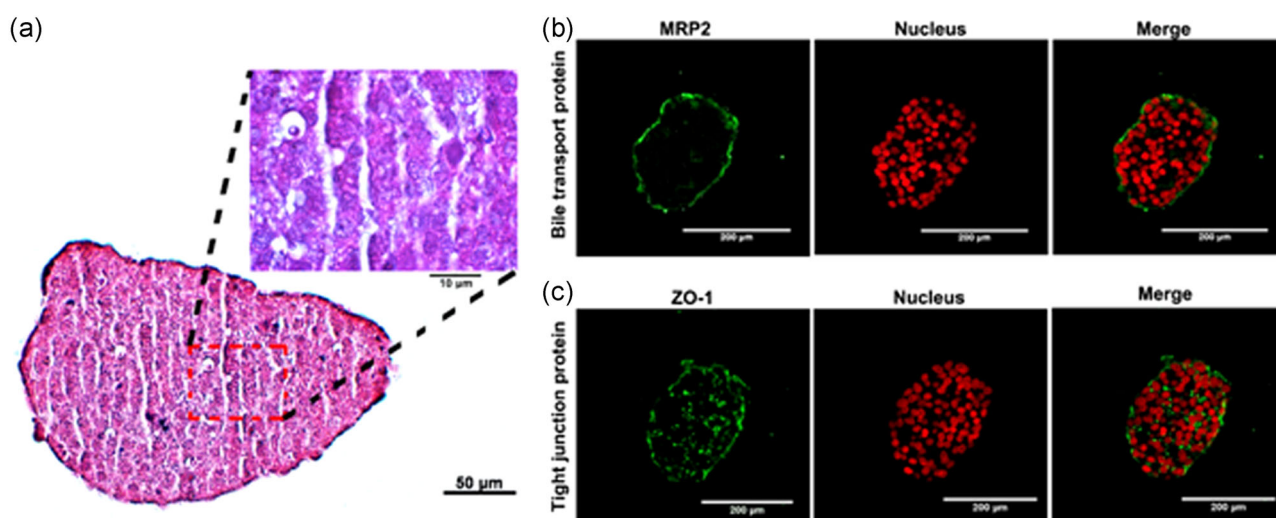


FIGURE 6 Histomorphological and immunohistochemical analysis of LPMTs. (a) Bright-field image of H&E stained microtissue showing hepatic plate-like cellular arrangement. Scale bar = 50 μm. (b, c) Confocal images showing the immunofluorescent stained LMPTs positive (green) for hepatocyte polarization markers MRP2 and ZO1. Cell nuclei were counterstained with PI (red). Scale bar = 200 μm. H&E, hematoxylin and eosin; LPMT, liver parenchymal microtissue; MRP2, Multidrug Resistance Protein 2; PI, propidium iodide; ZO1, Zonula Occludens 1

present *in vivo*. Immunofluorescence staining of LPMTs was performed to visualize the expression of hepatocyte polarization markers MRP2 (apical) and ZO1 (basolateral). Expression of MRP2 was primarily confined to the periphery of the microtissue with less abundance within (Figure 6b). However, positive staining for the protein ZO1 throughout the microtissue indicated the formation of tight intercellular junctions required for epithelialization (Figure 6c).

3.4 | LPMT sandwich culture

3.4.1 | Synthesis and characterization of GelMA

The overall percent yield of GelMA synthesis was 70% and an illustration of GelMA polymer is shown in Figure 2Sa. The ^1H NMR spectrum of the GelMA (Figure 2Sb) confirmed the functionalization of gelatin by the methacrylamide group. Resonance signals at 2, 5.5, and 5.7 ppm in GelMA correspond to methyl protons of methacrylamide grafts and acrylic protons of methacrylamide grafts on hydroxyl lysine and lysine, respectively. The signal at 3.09 ppm corresponding to methylene protons of unreacted lysine was lower in GelMA than gelatin.

The porosity of hydrogel is a detrimental factor that influences the diffusion of nutrients and metabolites, affecting cellular viability and functions. SEM images (Figure 2Sc) of the lyophilized GelMA showed a porous network of polymer with smooth walls separating the pores. Micro-CT analysis provided the cross-sectional images of lyophilized GelMA (Figure 2Sd) depicting porous internal structure. Quantitative morphometric analysis yielded a PV/TV ratio of 0.3867 and the percentage porosity was determined to be 61.33%. The pore diameter of the hydrogel ranged from 10 to 300 μm . The percentage pore volume occupied by pores having diameter 10–100, 100–200, and 200–300 μm was 67.2%, 29.5%, and 3.2%, respectively (Figure 2Se).

3.4.2 | LPMTs in GelMA and alginate sandwich culture

The suspension and sandwich culture of LPMTs is illustrated in Figures 7a,b. The LPMTs sandwiched in GelMA culture ensured immobilization of LPMTs, and the microtissues maintained an intact structure without any gross changes in size over 15 days (Figure 7c). The LPMTs in alginate showed irregular edges but with intact structure (Figure 7d). On the contrary, microtissues in suspension culture exhibited fusion and agglomeration (Figure 7e).

3.4.3 | Functional analysis of sandwich-LPMTs

Further, to understand the influence of extracellular matrix on the microtissue function, we compared the basic liver functionality of both groups. Liver-specific functions such as protein synthesis,

ammonia detoxification, and phase I metabolism-related enzyme activity in the LPMTs in sandwich culture were analyzed for 15 days. Albumin secreted into the culture medium represented the protein synthesis function of the liver. Though the albumin synthesis in sandwich-LPMTs on day 1 was similar to LPMTs in suspension, the synthesis increased over time. It was significantly higher than LPMTs in suspension on days 7 and 15 (Figure 7f). The detoxification ability of microtissues was assessed by the estimation of urea secreted into the culture medium (Figure 7g). Similar to the albumin secretion, urea secreted by sandwich-LPMTs increased over time and was significantly higher on days 7 and 15. In contrast, LPMTs in suspension showed a significant decrease in urea secretion. Cytochrome P450 3A4 monooxygenases are the liver's major enzymes involved in the phase I metabolism of xenobiotics. The enzyme activity of CYP3A4 increased significantly over time in sandwich-LPMTs compared with LPMTs in suspension, where the activity was substantially less throughout the culture (Figure 7h).

The organ-specific functions of LPMTs sandwiched on GelMA hydrogel (5% wt/v) were compared with LPMTs sandwiched in the same concentration of sodium alginate hydrogel. The albumin synthesis significantly increased in both sandwich systems from day 1 to 7 and was stable till the 15th day. However, the total amount of albumin synthesized in GelMA sw-LPMTs was significantly higher than that in Alginate sw-LPMTs throughout the culture period (Figure 8a). The urea synthesis significant linear increase in both the groups from day 1 to 15 while the total amount of urea synthesized in GelMA sw-LPMTs was always significantly higher than that in Alginate sw-LPMTs (Figure 8b). The CYP3A4 enzyme activity was maintained in GelMA sw-LPMTs, whereas the enzyme activity in Alginate sw-LPMTs significantly decreased from day 1 to 15 of culture (Figure 8c) indicating enhanced organ-specific functions of LPMTs in GelMA sandwich.

4 | DISCUSSION

Three-dimensional microtissues can serve as functional building blocks for tissue engineering as they mimic the physiological features *in vitro* (Mironov et al., 2009; Okudaira et al., 2016). Recent advances in the 3D culture of liver cells have shown that microtissues can be used as microscale tissue analogs to study cell–cell interaction, disease mechanism, and tissue response to drugs (Hurrell et al., 2020). Hanging-drop culture is the most straightforward and reproducible method for fabricating homogeneous microtissues (Pampaloni et al., 2007). It is a scaffold-free method wherein the cells self-assemble mimicking events during organogenesis (Achilli et al., 2012; Moscona & Moscona, 1952). At present commercialized hanging-drop culture plates (e.g., GravityPLUS plate, InSphero and Perfecta3D plate, 3D Biomatrix) are available in the standard microplate format. These are expensive (~100 US\$ per plate) culture wares, and the number of microtissues generated is limited. To scale up the production of microtissues, a cost-effective alternative has to be implemented that is precise and reproducible.

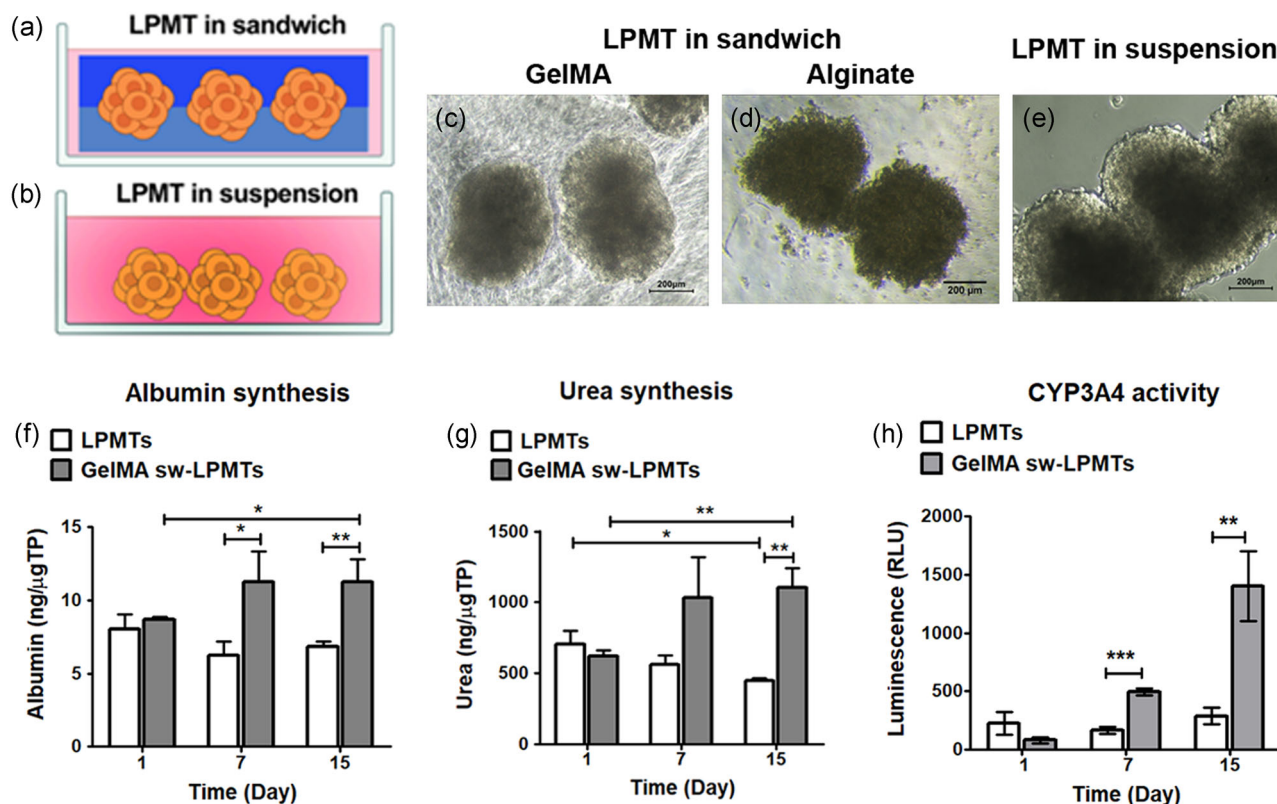


FIGURE 7 Comparison of the structural and functional integrity of LPMTs in the sandwich and suspension culture. (a, b) Schematic of the microtissue sandwich culture and suspension culture. Phase-contrast images of the microtissues sandwiched in (c) GelMA and (d) Alginate and (e) LPMTs in suspension. (f) Graph showing an increasing level of albumin secretion by sandwich-LPMTs compared with LPMTs. (g) Urea secreted by GelMA sandwich-LPMTs increased while that of LPMTs decreased. (h) CYP3A4 enzyme activity increased significantly sandwich-cultured LPMTs. Luminescence intensity correlates with the amount of luminescent product produced by the active enzyme (a pooled sample of 210 microtissues was analyzed in triplicates for all the assays, * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$). Scale bar = 200 μm . GelMA, gelatin methacrylamide; LPMT, liver parenchymal microtissue; sw-LPMT, sandwich-cultured LPMT

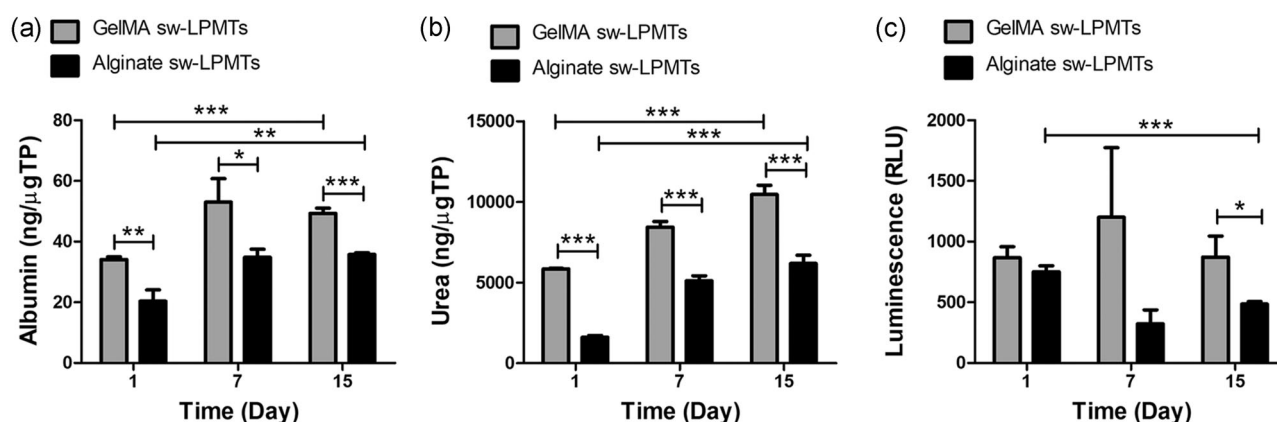


FIGURE 8 Comparison of the functional capabilities of LPMTs in the GelMA and Alginate sandwich culture. (a) Graph showing the enhanced level of albumin secretion by GelMA sandwich-cultured LPMTs compared with Alginate sandwich-cultured LPMTs. (b) Graph showing the enhanced level of urea secretion by GelMA sandwich-cultured LPMTs compared with Alginate sandwich-cultured LPMTs. (c) CYP3A4 enzyme activity is maintained in GelMA sandwich-cultured LPMTs while the activity decreases in Alginate sandwich-cultured LPMTs (a pooled sample of 210 microtissues was analyzed in triplicates for all the assays, * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$). GelMA, gelatin methacrylamide; LPMT, liver parenchymal microtissue; sw-LPMT, sandwich-cultured LPMT

In this study, three key aspects related to microtissue production and culture were addressed: (1) fabrication of a cost-effective and customizable hanging-drop culture platform, (2) microvalve-based bioprinting to generate cell-laden hanging drops in a high-throughput manner, and (3) preparation of a microtissue sandwich culture model for preserving structural–functional integrity of LPMTs *in vitro*.

The hanging-drop culture platform works in conjunction with the microvalve-based printing technology for the mass production of 3D microtissues. The substrate used for holding the drops was transparent thin PET sheets. It has been previously reported that PET sheets can serve as carriers for cell constructs (Joseph et al., 2011). The advantage of the PET sheet is its lightweight, optical clarity, and high strength-to-weight ratio. A custom-designed culture chamber that can accommodate PET carriers with hanging drops was designed. The laboratory level cost estimate for the fabrication of reusable HdCC was approximately 30 US\$, whereas PET culture substrates required for one HdCC unit is only 100 center coins. Both the chamber and the PET sheets can be sterilized using alcohol wiping, ultraviolet irradiation or, most efficiently, by ETO sterilization. Taken together, the total cost of our culture system was many times lesser than commercially available hanging-drop culture plates.

The manual dispensing of cell suspension for microtissue production by hanging-drop method is labor-intensive and has to be replaced with an automated liquid dispensing technology. There are reports on automated liquid handling systems for dispensing cell suspension into a specially designed hanging-drop multiwell plate (Drewitz et al., 2011; Tung et al., 2011). Recently, a drop-on-demand printing technique has been used to generate cell-laden hanging-droplets for producing 3D microtissues. Faulkner-Jones et al. (2013) first demonstrated the application of a valve-based cell printer to produce human embryonic stem cell spheroid aggregates. In their study, a dual nanoliter dispensing system was used to generate the hanging-drop array.

Similarly, Gutzweiler et al. (2017) reported the large-scale production of human umbilical vein endothelial cells spheroids using a disposable, electromagnetic dispensing valve. In this study, we utilized the microvalve-based bioprinting technique for preparing an array of cell-laden hanging droplets on the PET substrate. Hepatocarcinoma cell line (HepG2) was used for the *in vitro* experiments due to its unlimited life span and widely used as an alternative to primary hepatocytes (Gómez-Lechón et al., 2014). Although the metabolic representation of cytochrome P450 by HepG2 cells in conventional monolayer culture is deficient, 3D microtissues exhibit enhanced enzyme activity (Ramaiahgari et al., 2014). A high-precision microvalve was mounted on the printhead of a 3D bioprinter to dispense the droplets, and the parameters for the bioprinting were optimized. Microvalve-based bioprinting was proved to be an efficient method for the high-throughput production of microtissues compared with the manual dispensing system. The efficiency of microtissue formation by bioprinting was greater than 95%,

whereas it was only 60% in the manual method. However, due to the low viscosity of cell suspension, the cells tend to settle down during the printing, resulting in the variable number of cells per droplet (Faulkner-Jones et al., 2013). The cell sedimentation was prevented by short-time agitation of cell suspension using a cell-friendly agitator before each print job. The data presented in Figure 3e clearly show the uniformity in the size of the microtissues with agitation due to equally distributed cell number. Such improvisations in the process methodology increased the efficiency of our printing technique and yielded uniform microtissues. The efficiency of harvesting the microtissues from the substrate is also a detrimental factor in assessing the reliability of high-throughput methods. Gutzweiler et al. (2017) reported a high efficiency of spheroid formation (~96%) in hanging droplets, while the efficiency of harvesting the microtissues was significantly less (~54%). The hydrophilic nature of PET substrate allowed an easy collection of microtissues with a harvesting efficiency greater than 90%. Earlier reports have shown the advantages of an integrated approach consisting of automated cell dispensing with hanging-drop culture platforms. Popova et al. (2019) have developed a highly miniaturized single spheroid hanging-droplet microarray based on hydrophilic–superhydrophobic patterning to form tumor spheroids. However, it involves a series of complex fabrication steps for developing the droplet microarray device that can accommodate 588 spheroids in contrast to our HdCC that can yield 1600 spheroids at one go.

An inherent limitation of microtissues is the formation of hypoxic core with the increase in size, due to the limited diffusion of oxygen and nutrients (Asthana & Kisaalita, 2012). Since the size of the microtissue is directly proportional to the cell number dispensed per droplet, the size versus cell number was analyzed. As shown in Figure 4a, at the end of 3 days of initiating hanging-drop culture, the microtissue size showed a linear relationship to the cell number. When proliferating cell lines are used for microtissue culture, the cell multiplication increases cell number and thereby the size of LPMTs over the days. Hence lower cell number is preferred for seeding over high cell number to avoid necrotic core formation as showed by Gaskell et al. (2016). We opted for 500 cells per droplet as the optimum cell concentration because the radius of the spherical microtissue formed was close to 150 μm which was well within the range of diffusion limit. The viability staining ensured the absence of necrotic core and consistent levels of ATP production within the LPMTs over 2 weeks (Figure 5a,b).

One of the liver's primary functions is to assist in glucose homeostasis by regulating the interconversion of glucose and glycogen. Glucose uptake and glucose conversion to glucose-6-phosphate in hepatocyte are partly controlled by the hormone insulin (Iozzo et al., 2003). The sensitivity of LPMTs to insulin was analyzed by quantifying the uptake of labeled glucose after a short

period of starvation. Insulin-treated LPMTs showed a significant increase in glucose uptake, indicating an active response towards hormonal stimuli.

Earlier studies have shown that the microtissues preserve the native cell morphology and better cell-to-cell interactions compared with conventional monolayer culture (Peshwa et al., 1996; van Zijl & Mikulits, 2010). To characterize the structural features of LPMTs, we examined the stained sections. To our knowledge, this is the first report that has shown a typical hepatic plate-like cellular arrangement within microtissues developed by hanging-drop culture. Another essential structural characteristic that we have analyzed was the expression of the hepatocyte polarization markers MRP2 and ZO1. It has been reported that HepG2 spheroids formed on a PDMS mold expressed MRP2 and ZO1 when encapsulated in GelMA and are useful for the liver on a chip platform (Bhise et al., 2016). Expression of MRP2 and ZO1 in LPMTs generated using novel HdCC confirms that the cells are structurally polarized with epithelial characteristics.

Lack of control over the size and shape of microtissues and the need for pooling of microtissues to yield sufficient analyte for detection, hampers their use for high-throughput screening assays. Immobilization of microtissues within a definite space using a suitable biomatrix can be a solution. Earlier studies on collagen sandwich culture of hepatocytes have proven the positive impact of extracellular matrix on hepatocyte survival and functions (Dunn et al., 1989). Pinkse et al. (2005) have shown that the interactions between arginylglycylaspartic acid (RGD) peptides and integrins confer survival benefits to hepatocytes via ILK-pAkt pathway activation. In our study, we have chosen GelMA, a functionalized gelatin derivate that is rich in RGD peptides. Hence, it is expected that the GelMA sandwich culture will provide in vivo-like extracellular cues to the microtissues. First, we studied the advantages of GelMA as a biomatrix for enhancement of LPMT function in 10% GelMA sandwich and compared with LPMTs in suspension. Further to investigate the specific role of GelMA as a better biomatrix for LPMTs, we compared the organ-specific functions of GelMA sw-LPMTs with Alginate sw-LPMTs. Alginate is a polysaccharide-based polymer capable of forming hydrogels by ionic crosslinking. It is considered as an important biomaterial used for a wide range of applications like tissue engineering and drug delivery (Lan et al., 2010). However, the native form of the polymer lacks bioactive molecules present in the native extracellular matrix and has limited long-term stability (K. Y. Lee & Mooney, 2012). Hence, alginate hydrogel was chosen as a control for evaluating our GelMA hydrogel. However, it was difficult to prepare sandwich culture with 10% (w/v) alginate solution. Hence, we used 5% (w/v) of GelMA and Alginate for this experiment. Higher levels of liver functions in GelMA sw-LPMTs substantiated the fact that GelMA was a better biomatrix for enhancing organ-specific functions of LPMTs. Another critical factor that influences the diffusion of nutrients and metabolites across the matrix is its porosity. The morphological

characterization of GelMA hydrogel has proved that it is highly porous. The enhanced liver-specific functionalities of LPMTs in GelMA sandwich culture also support the fact that the mass transfer was not affected due to immobilization within the hydrogel.

Conclusively, a proof-of-concept of high-throughput production of LPMTs and their functional enrichment in vitro has been demonstrated successfully. However, a detailed investigation regarding the molecular changes that lead to the enhanced liver functions in sandwich culture has not been done. Our future research will focus on developing microtissues using primary hepatocytes and non-parenchymal cells to represent liver physiology better. An investigation at the molecular level will be carried out to establish the microtissue sandwich model further.

5 | CONCLUSIONS

A high-throughput method for LPMT production was developed using a microvalve-based bioprinting technique with a novel hanging-drop culture device. The microvalve-based bioprinting technique offered a high degree of miniaturization in cell number, droplet volume, and substrate area without compromising microtissue formation and viability. The simple design of the HdCC enabled the easy fabrication of the culture device at a significantly lower cost, and the use of PET substrate offered high tissue harvesting efficiency. This culture device can be quickly adopted in laboratories and industries with an automated liquid dispensing system. The microtissue sandwich culture preserved the structural integrity of LPMTs and enhanced the liver-specific functions without affecting the mass transfer of nutrients. The proposed methodology and the sandwich culture model will find potential application in liver tissue engineering and drug testing.

ACKNOWLEDGMENTS

Anil Kumar P. R. acknowledges the fund of Technical Research Center, SCTIMST (TRC/8141/PSN) received from the Department of Science and Technology, Government of India. Roopesh R. Pai thanks SCTIMST for the Ph.D. Fellowship. Senthilkumar Muthusamy is grateful to the Department of Biotechnology (DBT), Government of India for the research grant (BT/RLF/Re-entry/48/2016) with Ramalingaswami Fellowship. We thank Dr. Ganesh Kadasoor (Olympus) for the technical help in confocal imaging, Mr. Ramesh Babu (Precision Fabrication Facility, SCTIMST) for the fabrication of Hanging-drop Culture Chamber, Dr. D. R. Deepu for the Micro-CT analysis, and Central Laboratory and Instrumentation Facility, Kerala University for NMR spectroscopy.

CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

DATA AVAILABILITY STATEMENT

Data are available on request from the authors.

ORCID

Ramesh Pai Roopesh  <https://orcid.org/0000-0003-0311-4399>
 Senthilkumar Muthusamy  <https://orcid.org/0000-0003-1603-9965>
 Shiny Velayudhan  <https://orcid.org/0000-0002-3906-5915>
 Arumugham Sabareeswaran  <https://orcid.org/0000-0002-9365-1106>
 Pallickaveedu RajanAsari Anil Kumar  <https://orcid.org/0000-0003-1610-7323>

REFERENCES

- Achilli, T.-M., Meyer, J., & Morgan, J. R. (2012). Advances in the formation, use and understanding of multi-cellular spheroids. *Expert Opinion on Biological Therapy*, 12, 1347–1360. <https://doi.org/10.1517/14712598.2012.707181>
- Asrani, S. K., Devarbhavi, H., Eaton, J., & Kamath, P. S. (2019). Burden of liver diseases in the world. *Journal of Hepatology*, 70, 151–171. <https://doi.org/10.1016/j.jhep.2018.09.014>
- Asthana, A., & Kisaalita, W. S. (2012). Microtissue size and hypoxia in HTS with 3D cultures. *Drug Discovery Today*, 17, 810–817.
- Berthiaume, F., Moghe, P. V., Toner, M., & Yarmush, M. L. (1996). Effect of extracellular matrix topology on cell structure, function, and physiological responsiveness: Hepatocytes cultured in a sandwich configuration. *FASEB Journal*, 10, 1471–1484. <https://doi.org/10.1096/fasebj.10.13.8940293>
- Bhise, N. S., Manoharan, V., Massa, S., Tamayol, A., Ghaderi, M., Miscuglio, M., Lang, Q., Shrike Zhang, Y., Shin, S. R., Calzone, G., Annabi, N., Shupe, T. D., Bishop, C. E., Atala, A., Dokmeci, M. R., & Khademhosseini, A. (2016). A liver-on-a-chip platform with bioprinted hepatic spheroids. *Biofabrication*, 8, 14101. <https://doi.org/10.1088/1758-5090/8/1/014101>
- Chang, T. T., & Hughes-Fulford, M. (2009). Monolayer and spheroid culture of human liver hepatocellular carcinoma cell line cells demonstrate distinct global gene expression patterns and functional phenotypes. *Tissue Engineering: Part A*, 15, 559–567. <https://doi.org/10.1089/ten.tea.2007.0434>
- Drewitz, M., Helbling, M., Fried, N., Bieri, M., Moritz, W., Lichtenberg, J., & Kelm, J. M. (2011). Towards automated production and drug sensitivity testing using scaffold-free spherical tumor microtissues. *Biotechnology Journal*, 6, 1488–1496.
- Du, Y., Han, R., Wen, F., Ng San, S. S., Xia, L., Wohland, T., Leo, H. L., & Yu, H. (2008). Synthetic sandwich culture of 3D hepatocyte monolayer. *Biomaterials*, 29, 290–301. <https://doi.org/10.1016/j.biomaterials.2007.09.016>
- Dunn, J. C. Y., Yarmush, M. L., Koebe, H. G., & Tompkins, R. G. (1989). Hepatocyte function and extracellular-matrix geometry—Long-term culture in a sandwich configuration. *FASEB Journal*, 3, 174–177.
- Faulkner-Jones, A., Greenhough, S., King, J. A., Gardner, J., Courtney, A., & Shu, W. (2013). Development of a valve-based cell printer for the formation of human embryonic stem cell spheroid aggregates. *Biofabrication*, 5, 1758–5082.
- Gaskell, H., Sharma, P., Colley, H., Murdoch, C., Williams, D., & Webb, S. (2016). Characterization of a functional C3A liver spheroid model. *Toxicology Research*, 5, 1053–1065. <https://doi.org/10.1039/c6tx00101g>
- Glicklis, R., Merchuk, J. C., & Cohen, S. (2004). Modeling mass transfer in hepatocyte spheroids via cell viability, spheroid size, and hepatocellular functions. *Biotechnology and Bioengineering*, 86, 672–680. <https://doi.org/10.1002/bit.20086>
- Gómez-Lechón, M. J., Tolosa, L., & Mt, D. (2014). Cell-based models to predict human hepatotoxicity of drugs. *Reviews in Toxicology*, 31, 149–156. <https://www.redalyc.org/articulo.oa?id=91932969007>
- Gutzweiler, L., Kartmann, S., Troendle, K., Benning, L., Finkenzeller, G., Zengerle, R., Koltay, P., Stark, G. B., & Zimmermann, S. (2017). Large scale production and controlled deposition of single HUVEC spheroids for bioprinting applications. *Biofabrication*, 9, 1758–5090.
- Haisler, W. L., Timm, D. M., Gage, J. A., Tseng, H., Killian, T. C., & Souza, G. R. (2013). Three-dimensional cell culturing by magnetic levitation. *Nature Protocols*, 8, 1940–1949. <https://doi.org/10.1038/nprot.2013.125> [pii].
- Hiemstra, S., Ramaiahgari, S. C., Wink, S., Callegaro, G., Coonen, M., Meerman, J., Jennen, D., van den Nieuwendijk, K., Dankers, A., Snoeys, J., de Bont, H., Price, L., & van de Water, B. (2019). High-throughput confocal imaging of differentiated 3D liver-like spheroid cellular stress response reporters for identification of drug-induced liver injury liability. *Archives of Toxicology*, 93, 2895–2911. <https://doi.org/10.1007/s00204-019-02552-0>
- Hurrell, T., Kastrinou-Lampou, V., Fardellas, A., Hendriks, D. F. G., Nordling, Å., Johansson, I., Baze, A., Parmentier, C., Richert, L., & Ingelman-Sundberg, M. (2020). Human liver spheroids as a model to study aetiology and treatment of hepatic fibrosis. *Cells*, 9, 964.
- Iozzo, P., Geisler, F., Oikonen, V., Mäki, M., Takala, T., Solin, O., Ferrannini, E., Knuuti, J., & Nuutila, P. (2003). Insulin stimulates liver glucose uptake in humans: An 18F-FDG PET study. *Journal of Nuclear Medicine*, 44, 682–689.
- Jin, Y., Kim, J., Lee, J. S., Min, S., Kim, S., Ahn, D.-H., Kim, Y.-G., & Cho, S.-W. (2018). Vascularized liver organoids generated using induced hepatic tissue and dynamic liver-specific microenvironment as a drug testing platform. *Advanced Functional Materials*, 28, 1801954. <https://doi.org/10.1002/adfm.201801954>
- Lan, S. F., Safiejko-Mroccka, B., & Starly, B. (2010). Long-term cultivation of HepG2 liver cells encapsulated in alginate hydrogels: A study of cell viability, morphology and drug metabolism. *Toxicology in Vitro*, 24, 1314–1323. <https://doi.org/10.1016/J.TIV.2010.02.015>
- Lee, B. H., Kim, M. H., Lee, J. H., Seliktar, D., Cho, N.-J., & Tan, L. P. (2015). Modulation of Huh7.5 spheroid formation and functionality using modified PEG-based hydrogels of different stiffness. *PLOS One*, 10, e0118123. <https://doi.org/10.1371/journal.pone.0118123>
- Lee, B. H., Shirahama, H., Cho, N.-J., & Tan, L. P. (2015). Efficient and controllable synthesis of highly substituted gelatin methacrylamide for mechanically stiff hydrogels. *RSC Advances*, 5(128), 106094–106097. <https://doi.org/10.1039/c5ra22028a>
- Lee, K. Y., & Mooney, D. J. (2012). Alginate: Properties and biomedical applications. *Progress in Polymer Science*, 37, 106–126. <https://doi.org/10.1016/J.PROGPOLYMSCI.2011.06.003>
- Liao, W., Wang, J., Xu, J., You, F., Pan, M., Xu, X., Weng, J., Han, X., Li, S., Li, Y., Liang, K., Peng, Q., & Gao, Y. (2019). High-throughput three-dimensional spheroid tumor model using a novel stamp-like tool. *Journal of Tissue Engineering*, 10, 204173141988918.
- Manley, P., & Lelkes, P. I. (2006). A novel real-time system to monitor cell aggregation and trajectories in rotating wall vessel bioreactors. *Journal of Biotechnology*, 125, 416–424. <https://doi.org/10.1016/j.jbiotec.2006.03.030>
- Martignoni, M., Groothuis, G. M. M., & de Kanter, R. (2006). Species differences between mouse, rat, dog, monkey and human CYP-mediated drug metabolism, inhibition and induction. *Expert Opinion on Drug Metabolism & Toxicology*, 2, 875–894. <https://doi.org/10.1517/17425255.2.6.875>
- Mironov, V., Visconti, R. P., Kasyanov, V., Forgacs, G., Drake, C. J., & Markwald, R. R. (2009). Organ printing: Tissue spheroids as building blocks. *Biomaterials*, 30, 2164–2174. <https://doi.org/10.1016/j.biomaterials.2008.12.084>
- Moscona, A., & Moscona, H. (1952). The dissociation and aggregation of cells from organ rudiments of the early chick embryo. *Journal of Anatomy*, 86, 287–301. <https://pubmed.ncbi.nlm.nih.gov/12980879>

- Nguyen, D. G., Funk, J., Robbins, J. B., Crogan-Grundy, C., Presnell, S. C., Singer, T., & Roth, A. B. (2016). Bioprinted 3D primary liver tissues allow assessment of organ-level response to clinical drug induced toxicity in vitro. *PLOS One*, 11, e0158674. <https://doi.org/10.1371/journal.pone.0158674>
- Nithya, J., Anil Kumar, P. R., Tilak, P., Leena, J., Sreenivasan, K., Kumary, T. V. (2011). Intelligent thermoresponsive substrate from modified overhead projection sheet as a tool for construction and support of cell sheets In vitro. *Tissue Engineering Part C: Methods*, 17(2), 181–191. <http://doi.org/10.1089/ten.tec.2009.0783>
- Okudaira, T., Amimoto, N., Mizumoto, H., & Kajiwar, T. (2016). Formation of three-dimensional hepatic tissue by the bottom-up method using spheroids. *Journal of Bioscience and Bioengineering*, 122, 213–218.
- Okuyama, T., Yamazoe, H., Mochizuki, N., Khademhosseini, A., Suzuki, H., & Fukuda, J. (2010). Preparation of arrays of cell spheroids and spheroid-monolayer cocultures within a microfluidic device. *Journal of Bioscience and Bioengineering*, 110, 572–576.
- Pampaloni, F., Reynaud, E. G., & Stelzer, E. H. K. (2007). The third dimension bridges the gap between cell culture and live tissue. *Nature Reviews Molecular Cell Biology*, 8, 839–845. <https://doi.org/10.1038/nrm2236>
- Peshwa, M. V., Wu, F. J., Sharp, H. L., Cerra, F. B., & Hu, W.-S. (1996). Mechanistics of formation and ultrastructural evaluation of hepatocyte spheroids. *In Vitro Cellular & Developmental Biology—Animal*, 32, 197. <https://doi.org/10.1007/bf02722946>
- Pinkse, G. G., Jiawan-Lalai, R., Bruijn, J. A., & de Heer, E. (2005). RGD peptides confer survival to hepatocytes via the beta1-integrin-ILK-pAkt pathway. *Journal of Hepatology*, 42, 87–93. S0168-8278(04)00432-5 [pii]. <https://doi.org/10.1016/j.jhep.2004.09.010>
- Polonio-Alcalá, E., Rabionet, M., Gallardo, X., Angelats, D., Ciurana, J., Ruiz-Martínez, S., Puig, T., & Electrospun, P. L. A. (2019). Scaffolds for three-dimensional triple-negative breast cancer cell culture. *Polymers (Basel)*, 11(5), 916.
- Popova, A. A., Tronser, T., Demir, K., Haitz, P., Kuodyte, K., Starkuviene, V., Wajda, P., Levkin, P. A., Popova, A. A., Tronser, T., Demir, K., Levkin, P. A., Kuodyte, K., Starkuviene, V., & Wajda, P. (2019). Facile one step formation and screening of tumor spheroids using droplet-microarray platform. *Small*, 15, 1901299. <https://doi.org/10.1002/SMLL.201901299>
- Ramaiahgari, S. C., den Braver, M. W., Herpers, B., Terpstra, V., Commandeur, J. N. M., van de Water, B., & Price, L. S. (2014). A 3D in vitro model of differentiated HepG2 cell spheroids with improved liver-like properties for repeated dose high-throughput toxicity studies. *Archives of Toxicology*, 88, 1083–1095. <https://doi.org/10.1007/s00204-014-1215-9>
- Shri, M., Agrawal, H., Rani, P., Singh, D., & Onteru, S. K. (2017). Hanging drop, a best three-dimensional (3D) culture method for primary buffalo and sheep hepatocytes. *Scientific Reports*, 7, 17–1355.
- Tung, Y.-C., Hsiao, A. Y., Allen, S. G., Torisawa, Y., Ho, M., & Takayama, S. (2011). High-throughput 3D spheroid culture and drug testing using a 384 hanging drop array. *Analyst*, 136, 473–478. <https://doi.org/10.1039/c0an00609b>
- Wang, D., Cheng, D., Guan, Y., & Zhang, Y. (2011). Thermoreversible hydrogel for in situ generation and release of HepG2 spheroids. *Biomacromolecules*, 12, 578–584.
- Watkins, P. B. (2011). Drug safety sciences and the bottleneck in drug development. *Clinical Pharmacology & Therapeutics*, 89, 788–790. <https://doi.org/10.1038/clpt.2011.63>
- Yang, H., Sun, L., Pang, Y., Hu, D., Xu, H., Mao, S., Peng, W., Wang, Y., Xu, Y., Zheng, Y. C., Du, S., Zhao, H., Chi, T., Lu, X., Sang, X., Zhong, S., Wang, X., Zhang, H., Huang, P., ... Mao, Y. (2021). Three-dimensional bioprinted hepatorganoids prolong survival of mice with liver failure. *Gut*, 70, 567–574. <https://doi.org/10.1136/gutjnl-2019-319960> [pii].
- van Zijl, F., & Mikulits, W. (2010). Hepatospheres: Three dimensional cell cultures resemble physiological conditions of the liver. *World Journal of Hepatology*, 2, 1–7. <https://doi.org/10.4254/wjh.v2.i1.1>

SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

How to cite this article: Roopesh, P. R., Muthusamy, S., Velayudhan, S., Sabareeswaran, A. & Anil Kumar, P. R. (2022). High-throughput production of liver parenchymal microtissues and enrichment of organ-specific functions in gelatin methacrylamide microenvironment. *Biotechnology and Bioengineering*, 119, 1018–1032. <https://doi.org/10.1002/bit.28010>